Bagging and Random Forest

Tuesday, February 27, 2024 9:03 PM

- ML
  - supervised
    - LR — Logistic Regression
    - KNN
    - DT → JIO } Interpret } Regression / Classification
      - Entropy
      - Gini
        - IG

- DT → Overfit → Train 1 , Test 0.7

How to improve the results

↓

Class → 47 → Feedback

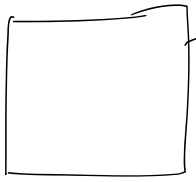


2 session → Today }

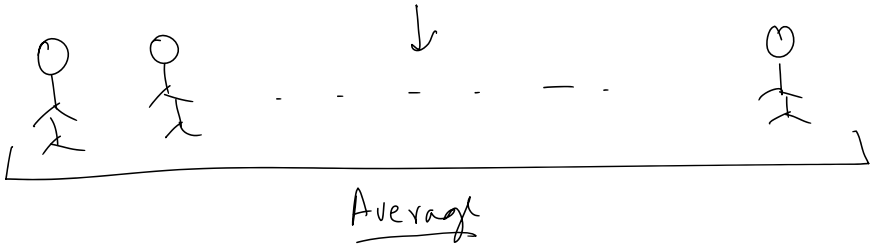
one ↓ 2 ← 10



10 — 10



D.T → 1 DT Tree — Leave / Not Leave

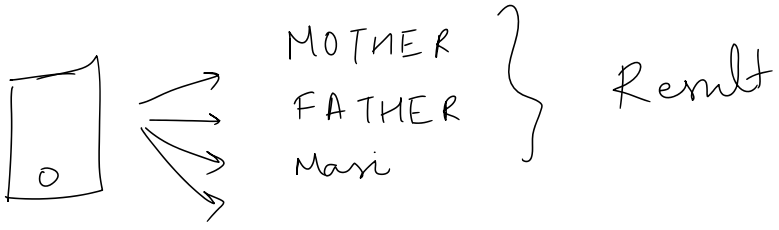


Ensemble

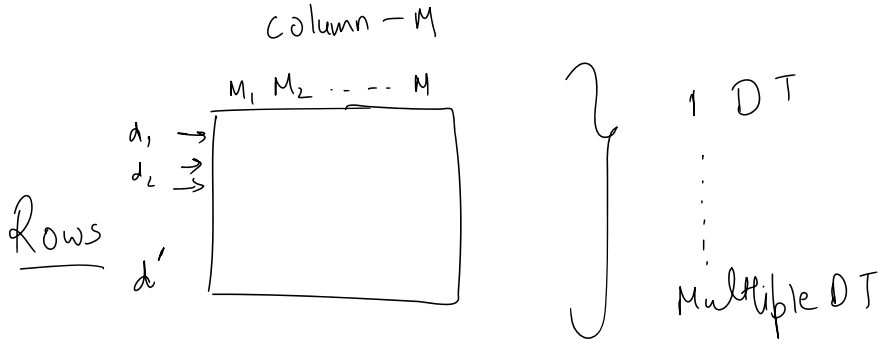
↓

Many base Learners + Combine

As - Different As possible



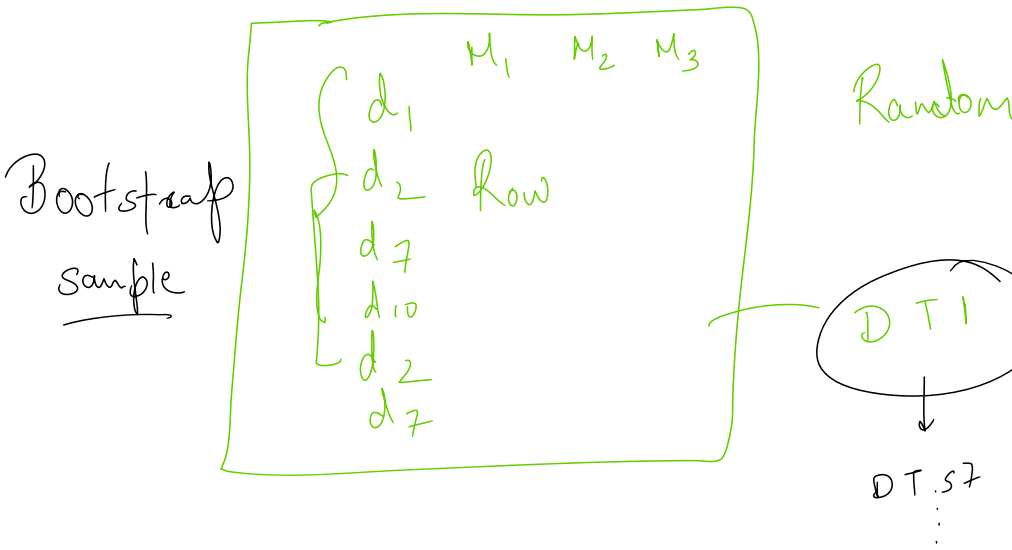
Bootstrap Aggregating



Duplicate

M1 M2 M3

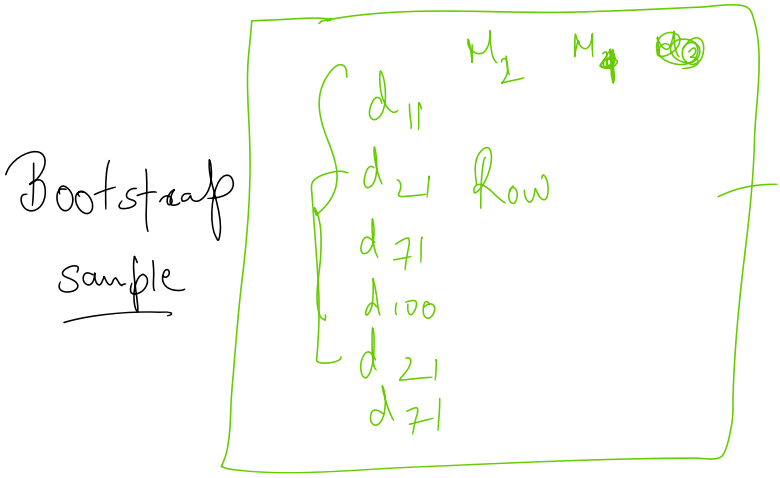
Random rows



Duplicate

Random → Random rows

→ Random Col



Sampling without Replacement

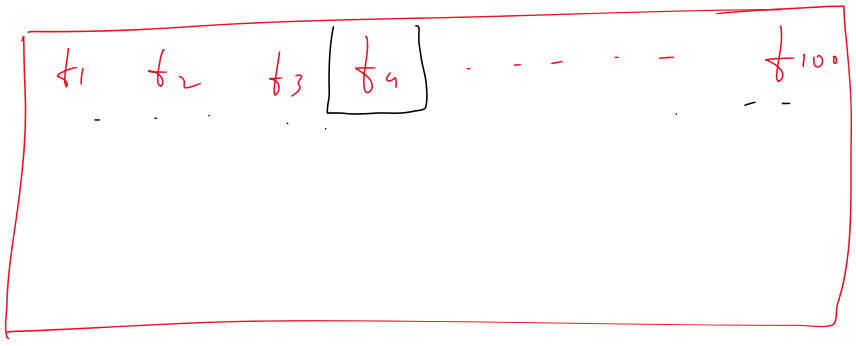
Random → Bootstrap

Forest → DT1 DT2

Bootstrap Random

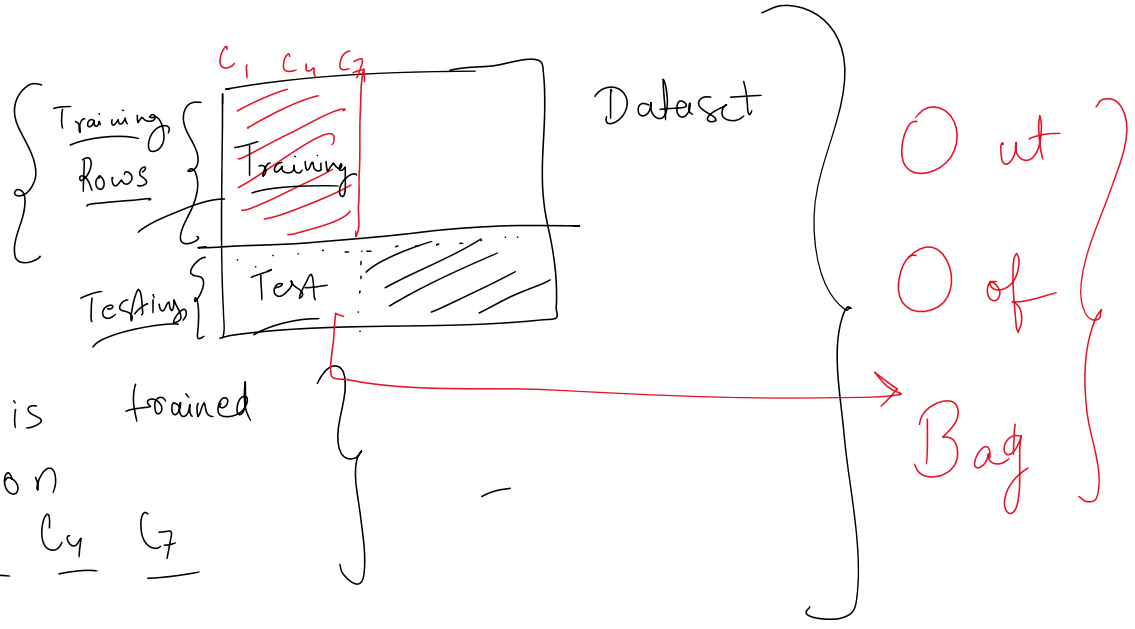
DT_1 DT_2

{ C5

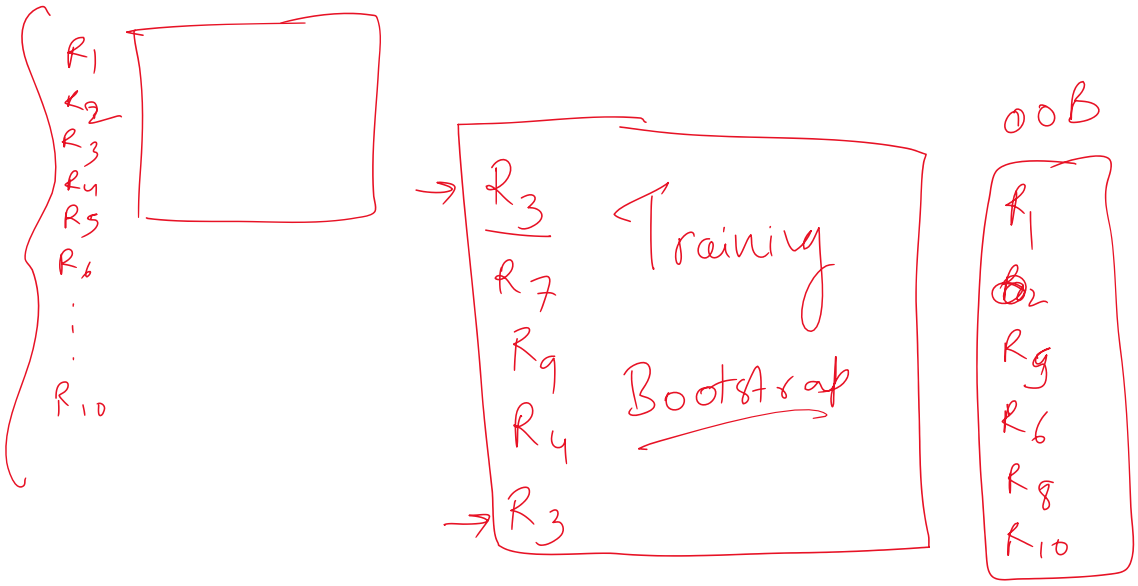


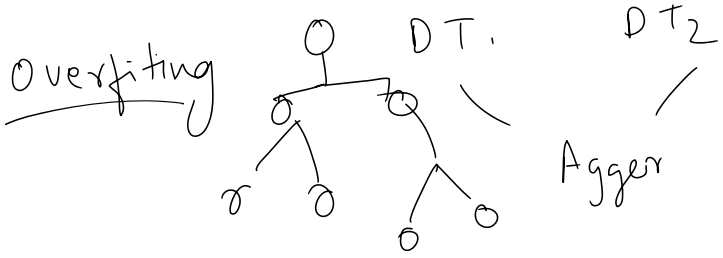
Overfit { x f3, x f7, x f8 } 3

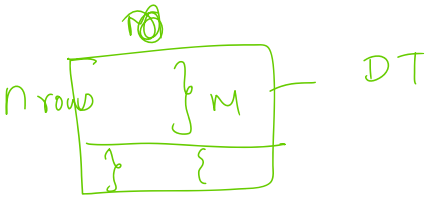
Ensemble



Random without Replacement







Overall performance of RF

Base learners are validated using → OOB

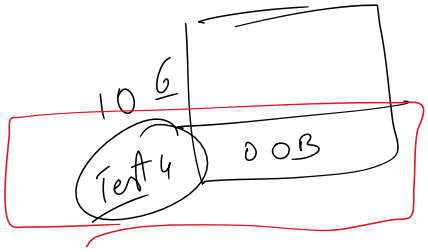
DT1 DT2

Base Learners

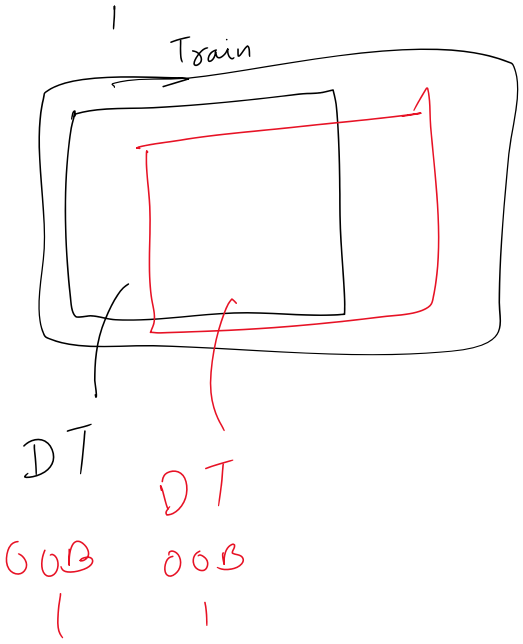
Test Data

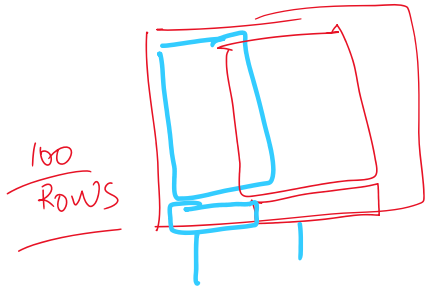
Hyperparmeter

No of DT



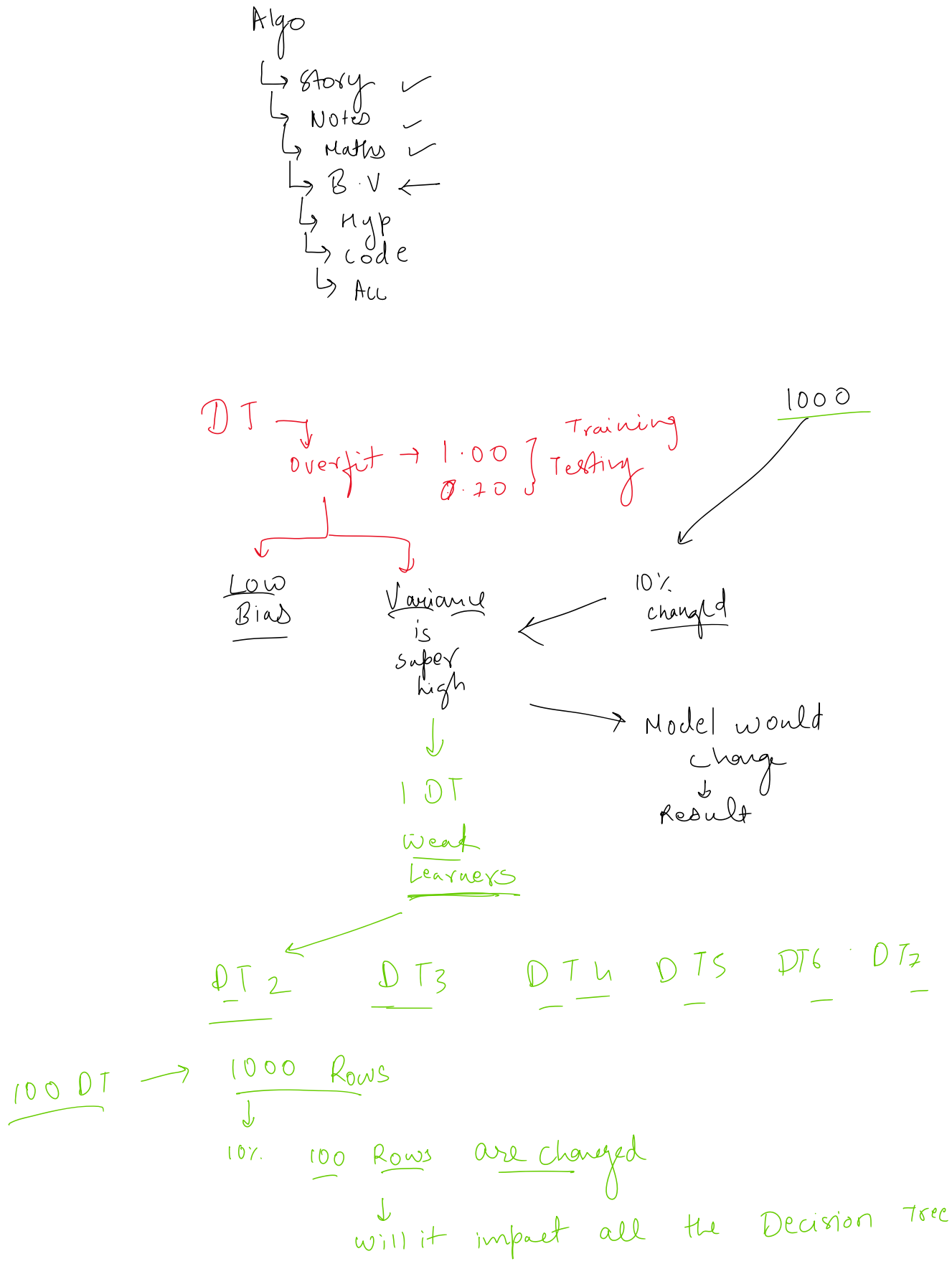
Small





Train

RF


Algo
Story
Notes
Maths
B.V
Hyp
Code
All
DT
Overfit → 1.00 Training
0.70 Testing
1000
Low Bias
Variance is super high
10% changed
Model would change
Result
1 DT
weak Learners
DT2
DT3
DT4
DT5
DT6
DT7
100 DT
1000 Rows
10% 100 Rows are changed
will it impact all the Decision Tree

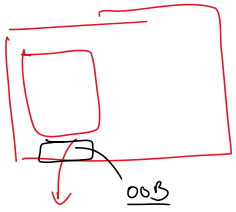
Bootstrap Aggregation

Bagging — RF

Random Sample

Overfit on one Model — Variance

100 Model



DT — Training

Parallel

Sample — DT_1 ← 70

Sample — DT_2 ← 80 DT

Sample — DT_3 ← 50

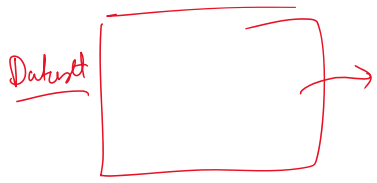
Sample — DT_4 ← 70

Agg → Pre

Rows → with Replacement (Duplicate)

Col → without Replacement (No Duplicate)

Col → without Replacement (No Duplicate)

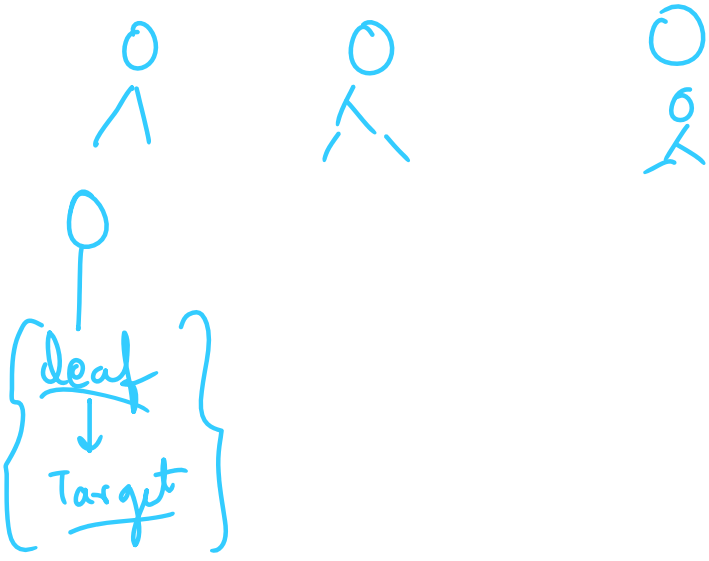


$D_1 \rightarrow$ R_1 R_2 R_7 R_{10} — C_1 C_7 C_3

$D_2 \rightarrow$ R_1 R_2 R_2 R_7 R_1 — C_1 C_2 C_7 C_1 C_4 — NO

Tree

$D_1 \rightarrow$ R_1 R_2 R_2 R_7 R_{10} — C_1 C_3 C_7



- N estimator → No of DT

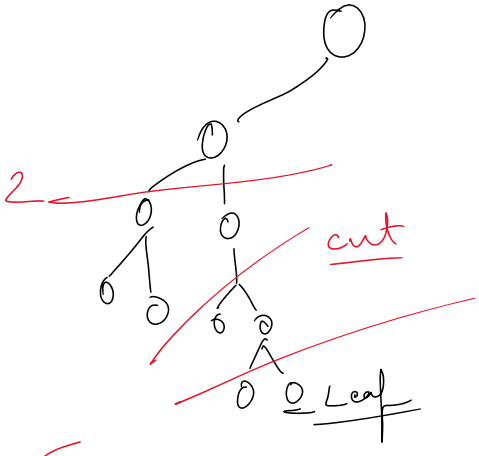
Max sample → Row [0.0, 1.0] RS
0.7

Max_features → sqrt √10 col → 3 col

Max_depth

Regularization → $\alpha \times |w^2|$

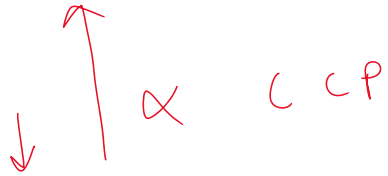
↳ Penalizing the coeff



Pruning
cut

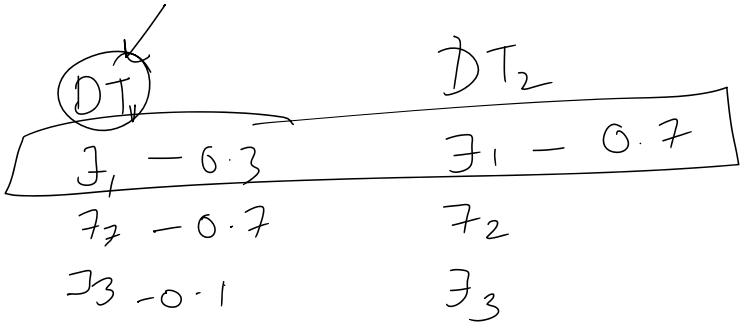
Cost complexity Pruning

CCP





Features – Imp



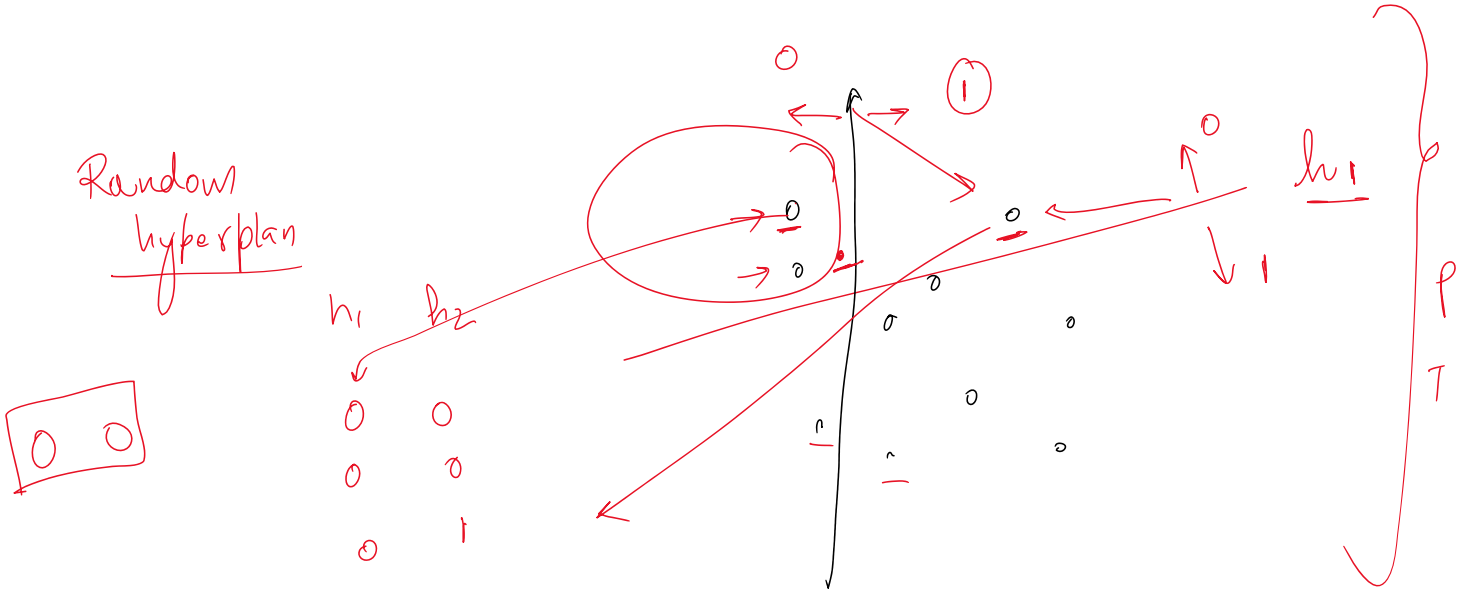
500 5000

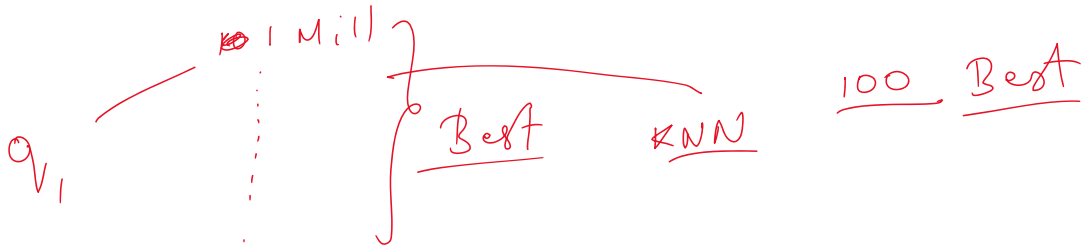
5000

LSH — Hashing } similar element

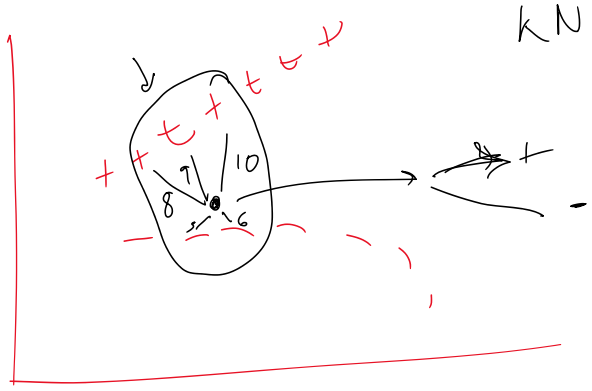
Locality — Sensitivity

Similar items will go into same bucket





KNN — k = 5



weighted KN

$$w = \frac{1}{\text{Distance}}$$

$$\frac{1}{8} + \frac{1}{9} + \frac{1}{10}$$

0.33

$$\frac{1}{5} + \frac{1}{6}$$

0.36

10x

GS { No of → 100, 200, 300, 400, 500, 700

{ Ram → 100, 200, 700, 400, 500, 700

Q. { 7 9